

Shweta U. Narkar, Ph.D.

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PROFESSIONAL SUMMARY

PhD-trained researcher working at the intersection of knowledge representation, scientific machine learning, and explainable AI, applied to domains where data schemas are undefined, and ground truth is expensive. Developed POLO (Prebiotic Origins of Life Ontology), the first formal ontology for experimental prebiotic chemistry, and a full-stack data ingestion framework enabling structured reasoning over 200+ hydrothermal experiments. Experienced building reproducible AI pipelines across computational and experimental teams, from problem formulation and data strategy through modeling and deployment. Skilled in OWL/RDF, SPARQL, RML mapping, Python-based ML, and SHAP-based explainability.

EXPERIENCE

AI/ML & Data Scientist

Feb 2026 – Present

Mirro.ai

Remote

- Developing a benchmarking pipeline in Python to evaluate speech emotion recognition model performance against labeled audio datasets.
- Analyzing end-to-end ML pipeline architecture across multiple services to identify optimization opportunities.
- Contributing to research direction for future model development and product initiatives.

Postdoctoral Research Associate

Aug 2025 – May 2026

Rensselaer Polytechnic Institute

Troy, NY

- Preparing manuscript for submission on POLO (Prebiotic Origins of Life Ontology), targeting the International Semantic Web Conference (ISWC).
- Collaborate with experimental teams to transform laboratory workflows into structured, ML-ready datasets for downstream modeling and analysis.
- Design and curate scientific datasets enabling reproducible data pipelines and predictive modeling.
- Mentoring 6 researchers across undergraduate, graduate, and high school levels in research methodology, data analysis, and scientific communication.

Machine Learning Research Intern

May 2020 – Sept 2020

IBM Research

Yorktown Heights, NY

- Designed and implemented Model LineUpper, a system for multi-level comparison of machine learning models enabling expert evaluation beyond aggregate performance metrics.
- Led a controlled user study with 14 professional data scientists evaluating 16 machine learning models, translating findings into actionable insights and recommendations for technical and non-technical stakeholders.
- Applied explainable AI techniques (SHAP-based feature importance) to analyze model behavior and support instance-level error diagnosis.
- Co-authored peer-reviewed research presented at the 2021 Intelligent User Interfaces Conference.

PhD Research Assistant, Applied ML & Data Science

May 2021 – May 2025

Rensselaer Polytechnic Institute

Troy, NY

- Developed POLO, the first formal ontology for experimental prebiotic chemistry, comprising 19 classes, 7 object properties, and 41 data properties, producing 12,000+ RDF triples via a declarative RML mapping pipeline and enabling structured annotation and reuse of 200+ experiments.
- Modeled experimental outcomes using machine learning (Random Forest, Gradient Boosting) on a curated dataset, achieving $R^2 = 0.734$ and informing the design of 24 follow-up experiments.
- Built a reusable data ingestion framework (Express.js, MongoDB) capturing 190+ experiments, enabling scalable dataset growth for downstream modeling and analysis.
- Collaborated with experimental scientists to analyze datasets and develop visualizations supporting hypothesis generation.

PhD Research Assistant, Scientific Data Systems

Jun 2018 – Apr 2021

Rensselaer Polytechnic Institute

Troy, NY

- Analyzed and visualized microbial locality datasets spanning 1000+ enzymes, supporting biochemical and geobiological research.

- Curated and standardized 10+ archival scientific datasets for the NASA-funded Earth First Origins project, improving data accessibility for downstream analysis.
- Developed a diabetes ontology as part of a graduate team project, applying knowledge representation principles to biomedical domain modeling.

SELECTED PUBLICATIONS

- Narkar et al. *Model LineUpper: Supporting Interactive Model Comparison at Multiple Levels for AutoML*. Proceedings of the 26th International Conference on Intelligent User Interfaces (IUI), 2021.
- Boujibar A. et al. (incl. Narkar S.U.) Cluster Analysis of Presolar Silicon Carbide Grains: Evaluation of Their Classification and Astrophysical Implications. *Astrophysical Journal Letters*, 2021.
- Invited Talk: Leveraging Computational and Semantic Technologies in Understanding Prebiotic Chemistry. Geological Society of America, 2023.
- Talk: Enabling Semantic Technologies and eScience for Exploring the Prebiotic Chemistry Experimental Landscape. AbSciCon, 2024.
- Narkar S.U., Riggi V.S., Rogers K.L., Hendler J.A. POLO: A Formal Ontology for Experimental Prebiotic Chemistry. *Manuscript in preparation*, target venue: International Semantic Web Conference (ISWC).
- Narkar S.U. et al. A Data Framework for Experimental Prebiotic Chemistry. *Manuscript in preparation*.
- Narkar S.U. et al. Analysis of the Experimental Landscape of Prebiotic Chemistry. *Manuscript in preparation*.

EDUCATION

Rensselaer Polytechnic Institute	Troy, NY
Ph.D. in Computer Science (Multidisciplinary: CS + Astrobiology)	May 2025
Rensselaer Polytechnic Institute	Troy, NY
M.S. in Data Science & Analytics	May 2019
University of Pune	India
B.E. in Computer Engineering	Aug 2017

- Developed a diabetic patient readmission prediction model using decision tree classification; selected as one of top 15 undergraduate seminars.

TECHNICAL SKILLS

Programming: Python, R

Machine Learning & Data Science: scikit-learn, pandas, NumPy, statistical modeling, model evaluation, PyTorch, SHAP

Data Systems & Tools: MongoDB, PostgreSQL, Git, Docker, Azure DevOps, Jupyter, GCP, GitHub

Visualization: Matplotlib, seaborn, ggplot2, D3.js

Other: Hugging Face Transformers, Ontology Engineering, Knowledge Graphs, OWL/RDF, RML Mapping, SPARQL, LLMs

LEADERSHIP & SERVICE

- Student Officer and Interim Secretary-Treasurer, Geoinformatics & Data Science Division, Geological Society of America.
- Teaching Assistant, Rensselaer Polytechnic Institute (CS I; Econometrics in Big Data with R).
- Mentor, Rensselaer Research Experience program supporting undergraduate research training.
- Recipient, RPI Founders Award of Excellence.